TED (15) – 4073 Reg. No. ………………

Revision – 2015 Signature. ………………

FOURTH SEM. CHEMICAL ENGINEERING (Model Qn.)

INORGANIC CHEMICAL TECHNOLOGY

Time: 3 hours

(Maximum marks : 100)

PART - A

(Maximum marks : 10)

Marks

I Answer the following questions in one or two sentences. Each question carries 2 marks

1 Indicate the anodic and cathodic reactions behind of caustic soda manufacture by membrane process

2 List four uses of chlorine

3 Define catalyst poisoning

4 Illustrate the optimum conditions for ammonia synthesis

5 List the principal raw materials for glass making

(5 X 2=10)

PART - B

(Maximum marks : 30)

II Answer any five of the following questions. Each question carries 6 marks

1 Explain ammonia recovery from manufacturing of soda ash

2 Draw membrane cell and label its parts

3 Illustrate the working of waste heat boilers

4 Compare the advantages and disadvantages of vanadium pentoxide and platinum

5 Draw a neat sketch and label the parts of an ammonia converter

6 Explain the manufacture of single super phosphate

7 Explain the working of rotory kiln (6 x 5 =30)

PART - C

Answer one full question from each unit

Each question carries 15 Marks

MODULE I

III	a	Support with a flow diagram explain modified Solvay process	8
	b	Explain the chlorine compression and liquefaction	7
		OR	
IV	a	Explain the manufacturing process of Aluminum Sulphate from Bauxite.	7
	b	Explain the process of manufacture of lime from limestone	8

MODULE II

V	a	Draw a neat diagram and label the parts of SO_2 converter	9
	b	Explain the sulphuric acid dehydrtion method of concentration of nitric acid	6
		OR	
VI	a	Explain the principles of DCDA process with flow sheet	9

	b	Draw a neat diagram and label the parts of HCl absorber	6
		MODULE III	
VII	a	Explain the manufacturing of ammonium sulphate with a neat flow diagram	8
	b	Summarise the kinetics and equilibria of the reaction involved in urea manufacturing	7
		OR	
VIII	a	Describe the manufacturing process of tripple super phosphate	8
	b	Draw a neat flow diagram of steam reforming process of ammonia	7
		MODULE IV	
IX	a	Explain the chloride process of manufacturing Titanium dioxide with flow sheet	10
	b	Water is added for the setting of cement'. Justify	5
		OR	
X	a	Explain the manfacture of sodium silicate	8
	b	Describe high alumina refractory brick	7